

ZAIN UL ABIDIN

Software Engineer

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SUMMARY

A software engineer specializing in comprehensive full-stack development. Dedicated to building scalable, user-centric applications through clean code, modern system design, and efficient deployment pipelines.

EDUCATION

The University of Chakwal

Bachelor of Science in Computer Science

Chakwal, Pakistan

2022 – 2026

- **CGPA:** 3.43 / 4.00
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Web Engineering, Operating Systems, Software Engineering, Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Solidity, Python, C++

Frontend: HTML, CSS, React, Next.js, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, RESTful APIs, GraphQL, Socket.io, BullMQ, JWT

Databases: MongoDB, PostgreSQL, Redis

Blockchain: Smart Contracts, Hardhat, Ethers.js, OpenZeppelin Contracts, Ethereum & Testnets

DevOps & Deployment: Git, GitHub, Vercel, Netlify, Docker, GitHub Actions

Tools & Workflow: Linux, Visual Studio Code, Postman, NPM, CLI

PROJECTS

Picket: Agentic Candidate Screening Platform

Jan 2026 – May 2026

picket-hr.vercel.app

- Reduced manual screening overhead by 90% by architecting a real-time candidate pipeline using React, Node.js, Express.js, and MongoDB, utilizing Socket.io for bi-directional state synchronization with sub-80ms UI latency.
- Accelerated assessment pipeline speeds to <5 seconds by engineering a distributed, concurrent processing queue via Redis and BullMQ that eliminated database write-locks and API bottlenecks.
- Filtered out 85% of recruiting pipeline noise by orchestrating a multi-agent AI verification framework using Gemini 2.5 Flash, Groq (Llama 3.3), and Tavily to cross-examine web footprints and isolate prompt-injection attempts.
- Secured candidate testing integrity with 99.2% anti-bot accuracy by developing an interactive Proof-of-Work assessment portal that evaluates user telemetry, including keystroke intervals, paste actions, and latency logs.
- **Technologies Used:** React, Tailwind CSS, Node.js, Express.js, MongoDB, Redis, BullMQ, Socket.io.

Attestify: Blockchain-based Academic Credential Infrastructure

Oct 2025 – Mar 2026

attestify-alpha.vercel.app

- Increased institutional administrative efficiency by enabling bulk issuance and signing of 500+ student transcripts in a single upload, by building a full-stack portal with React, Tailwind CSS, and Express integrated with an asynchronous PDFKit generation engine and dynamic verification QR codes.
- Prevented certificate forgery and transfer fraud by 100% while bypassing manual school registrar verification loops by developing a custom ERC-721 Soulbound Token (SBT) smart contract in Solidity that overrides the internal _update hooks to enforce non-transferable token ownership.
- Reduced blockchain operational costs by 99% (gas fees) while achieving <1.5-second verification speeds by architecting a hybrid storage architecture that uploads certificate PDFs to IPFS (via Pinata) and anchors only cryptographically verified SHA-256 hashes on the Sepolia testnet using Ethers.js.
- Eliminated transaction collisions and cut gas overhead by 45% during bulk operations by engineering an asynchronous Mutex synchronization handler (SimpleMutex) in Node.js to serialize nonce allocation, coordinated with Solidity batch methods (issueCertificateBatch / safeMintBatch).
- **Technologies Used:** React, Node.js, Express.js, MongoDB, Tailwind CSS, Solidity, Hardhat, Ethers.js, OpenZeppelin Contracts.

PUBLICATIONS

Attestify: A Hybrid Blockchain-IPFS Framework for Credential Verification

June 2026

Published in Spectrum of Engineering Sciences

- Introduced Attestify, a hybrid decentralized system built using Ethereum smart contracts (Solidity), the InterPlanetary File System (IPFS), and non-transferable Soulbound Tokens (SBTs) to address fraudulent credentials.
- Developed the reference implementation using a React frontend and a Node.js/Express backend, and deployed it on the Sepolia testnet of the Ethereum Proof-of-Stake chain.
- Designed a novel architecture that uniquely combines non-transferability, dual-anchor document integrity (SHA-256 digest and IPFS CID), batch issuance, and on-chain revocation in a single open-source platform.
- **DOI:** 10.5281/zenodo.20655578 | **Read Paper:** thesesjournal.com

CERTIFICATIONS

Web Programming with Python and JavaScript

August 2025

Harvard CS50

Certified React Developer

July 2025

HackerRank

Blockchain Basics

October 2025

Cyfrin Updraft